

Aaron Espinoza

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PUBLICATIONS

Enhance Multi-modal and Multi-center Whole Heart Segmentation Using Data Augmentation and Model Calibration. *CARE Workshop, MICCAI 2024*. doi.org/10.1007/978-3-031-87009-5_13

[Under review] Silent but Strong: Skip-Bottleneck Composition in Adversarial Face Impersonation. *Submitted to AAAI 2027*.

RESEARCH EXPERIENCE

Undergraduate Deep Learning Researcher

Texas State University

Aug. 2024 – Present

San Marcos, TX

- Research GAN-based adversarial perturbation generation against facial recognition systems, using hybrid U-Net generators to produce perturbations that transfer across models while remaining visually imperceptible.
- Achieved a 98.7% average impersonation success rate across standard benchmarks (LFW, CASIA-WebFace).
- Designed both targeted and untargeted attack formulations, trading off match-accuracy degradation against perceptual fidelity, and built the full PyTorch training and evaluation pipeline: preprocessing, adversarial training loop, and attack-success metrics.
- Sole undergraduate on the project; first-author paper under review at AAAI 2027.

AI Research Intern

Guangdong Provincial People's Hospital

May 2024 – Aug. 2024

Guangzhou, China

- Placed 1st globally in the MICCAI 2024 WHS++ whole heart segmentation challenge, competing against teams from research labs and industry.
- Built ensembled PyTorch CNN segmentation models for cardiac CT and MRI, reaching a 94% Dice score.
- Addressed generalizing across imaging modalities and hospital sites with targeted data augmentation and post-hoc model calibration.
- Co-authored the resulting workshop publication, working in a research team against a fixed challenge deadline.

ENGINEERING EXPERIENCE

Google Summer of Code Contributor

The FreeBSD Project

May 2025 – Sept. 2025

Remote

- Delivered a modular Rust framework for FreeBSD kernel device drivers with safe wrappers over C kernel objects, now used by other contributors.
- Wrote 11 automated ATF/Kyua tests covering stress, concurrency, error handling, and memory leaks, wired into CI — the same reproducibility discipline applied to systems code.
- Continue as an open-source contributor to the FreeBSD src tree (Sept. 2025 – Present). Writeup: gist.github.com/Acesp25 | Project video: youtu.be/y82-t1tDLWg

SELECTED PROJECTS

RustKLD

github.com/Acesp25/rustkld

Framework and CI test harness for FreeBSD kernel drivers written in Rust, with contributor documentation.

Rust, C, FreeBSD, ATF/Kyua.

Self-Hosted Compute and Storage

Self-hosted

One Ubuntu and two FreeBSD servers running QEMU VMs, bhyve VMs, and jails for experiment isolation, plus an NFS store reachable through WireGuard tunnels. Used to run training jobs and keep datasets versioned off a laptop.

QEMU, bhyve, jails, NFS, WireGuard, pf.

EDUCATION

Texas State University

B.S. Honors Computer Science; Minors in Applied Mathematics and Chinese

Expected Dec. 2026

San Marcos, TX

- GPA 3.98/4.00 overall, 4.00/4.00 College of Science & Engineering GPA. Dean's List/President's List, all semesters.
- Extracurriculars: Vice President, TXST Fencing Club (40+ members), TXST Esports Club CS2 Captain.
- Coursework: Linear Algebra, Probability & Statistics, Data Structures & Algorithms.

SKILLS

Machine Learning: PyTorch, TensorFlow, CNNs, U-Nets, GANs, adversarial attacks, image segmentation, ensembling, calibration, data augmentation

Languages: Python, C, Rust, Java, C++, Bash

Math: Linear algebra, probability and statistics, numerical methods (Applied Mathematics minor)

Tooling: Git, Linux, FreeBSD, Anaconda, Jupyter, VS Code, Vim, L^AT_EX, bhyve, jails, NFS